

Lemma 2.8, it follows that $\widetilde{K} \cap \widetilde{P}_1 \trianglelefteq \widetilde{P}_1$. As shown in Equation 3.1, the projection of y is given by $([\mu, \lambda][\mu, \lambda^2], 1)$, which lies in $\widetilde{K} \cap \widetilde{P}_1$. By Lemma 3.3, we have $P = \langle [\mu, \lambda], [\mu, \lambda][\mu, \lambda^2] \rangle$, implying that $\widetilde{P}_1/(\widetilde{K} \cap \widetilde{P}_1)$ is cyclic. Since P is perfect, it follows that $\widetilde{K} \cap \widetilde{P}_1 = \widetilde{P}_1$. Consequently, we conclude that $\widetilde{K} = \widetilde{P}_1 \times \widetilde{P}_3$. $\blacksquare$

To streamline the proofs of the subsequent lemmas, for any subgroup Q of P , we consistently denote by Q_i the corresponding subgroup in the i -th direct factor P_i and denote by Q^5 their direct product $Q_1 \times \dots \times Q_5$.

Lemma 3.5. *The graph Γ is a connected graph.*

Proof. Let P be a minimal counterexample for the statement $X \neq G$. Then for any nontrivial normal subgroup M of P , the quotient group P/M is perfect and generated by μM and λM . Let $L = M^5$. Then $X/L \cong (P/M) \wr A_5$, and by assumption, $X/L = GL/L$. Therefore, $X = GL$, and it follows that

$$N = GL \cap N = (G \cap N)L = KL.$$

We claim that $Z = Z(P)$, the center of P , is trivial. If $Z \neq 1$, From the discussion above, we have $N = Z^5 K$. Thus, $N = N' = (Z^5 K)' = K$, and hence $N = K$. Therefore, $X = G$, which contradicts that P is a counterexample. Hence, our claim holds.

Step 1: the projection of K on $P_1 \times P_2 \times P_3$ is surjective

Assume the projection of K on $P_1 \times P_2 \times P_3$ is not surjective. Let $U = P_1 \times P_2 \times P_3$ and $V = P_4 \times P_5$ and let $\widetilde{K}$ and $\widetilde{P}_i$ be the projections of K and P_i on U , where $1 \leq i \leq 3$. Then $\widetilde{K} = KV/V$, $\widetilde{P}_i = P_i V/V$ and $KV < N$. According to Lemma 3.4(ii) and Lemma 2.8, we have

$$(\widetilde{P}_1 \times \widetilde{P}_2)/(\widetilde{K} \cap \widetilde{P}_1 \times \widetilde{P}_2) \cong \widetilde{P}_3/(\widetilde{K} \cap \widetilde{P}_3).$$

This implies that $\widetilde{K} \cap (\widetilde{P}_1 \times \widetilde{P}_2) \neq 1$.

Suppose $\widetilde{K} \cap \widetilde{P}_1 \neq 1$. Since $\widetilde{K}$ projects surjectively onto $\widetilde{P}_2 \times \widetilde{P}_3$, it follows that $\widetilde{K} \cap \widetilde{P}_1$ is a nontrivial normal subgroup of $\widetilde{P}_1$. Thus, there exists a nontrivial normal subgroup D_1 of P_1 such that $D_1 \leq KV$. Since G transitively permutes $\{P_1, \dots, P_5\}$ under conjugation, we can find nontrivial normal subgroups $D_i \trianglelefteq P_i$ for $1 \leq i \leq 5$ such that G is transitive on $\{D_1, \dots, D_5\}$. Hence $\prod_{i=1}^5 D_i \trianglelefteq X$. Considering that the stabilizer of G on $\{P_4, P_5\}$ is transitive on $\{P_1, P_2, P_3\}$, we deduce that $D_1 \times D_2 \times D_3 \leq KV$. Then $(\prod_{i=1}^5 D_i) \leq KV$ as $D_4 \times D_5 \leq V$. Recall that $N = (\prod_{i=1}^5 D_i)K$. This implies that $(\prod_{i=1}^5 D_i)K \leq KV$, which contradicts $KV < N$.

Therefore, we must have $\widetilde{K} \cap \widetilde{P}_1 = 1$, and similarly, $\widetilde{K} \cap \widetilde{P}_2 = 1$. Since

$$(\widetilde{K} \cap (\widetilde{P}_1 \times \widetilde{P}_2)) \cap \widetilde{P}_1 = \widetilde{K} \cap \widetilde{P}_1 = 1,$$

and $\widetilde{K} \cap (\widetilde{P}_1 \times \widetilde{P}_2) \triangleleft \widetilde{P}_1 \times \widetilde{P}_2$, it follows that $\widetilde{K} \cap (\widetilde{P}_1 \times \widetilde{P}_2)$ centralizes $\widetilde{P}_1$. Recalling that $Z(\widetilde{P}_1) = 1$, the centralizer of $\widetilde{P}_1$ in $\widetilde{P}_1 \times \widetilde{P}_2$ is $\widetilde{P}_2$. Thus $\widetilde{K} \cap (\widetilde{P}_1 \times \widetilde{P}_2) \leq \widetilde{P}_2$. By